

EXPERIMENT-5

Write a program of Missionaries and Cannibals using python code

Aim : To solve the Missionaries and Cannibals problem using Python.

Procedure:

Represent the number of missionaries and cannibals on the left bank.

Move people across the river using a boat of capacity 2.

Ensure cannibals never outnumber missionaries on either bank.

Use Breadth-First Search (BFS) to find a valid sequence of moves.

Print the steps from the start state to the goal state.

Python Code:

```
from collections import deque

start = (3, 3, 1)    # (Missionaries, Cannibals, Boat)

goal = (0, 0, 0)

moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]

def valid(state):

    m, c, b = state

    if m < 0 or c < 0 or m > 3 or c > 3:

        return False

    if (m > 0 and m < c):

        return False

    if ((3-m) > 0 and (3-m) < (3-c)):

        return False

    return True
```

```

queue = deque([(start, [start])])

visited = set()

while queue:

    state, path = queue.popleft()

    if state == goal:

        print("Solution Path:")

        for step in path:

            print(step)

            break

    if state in visited:

        continue

    visited.add(state)

    m, c, b = state

    for dm, dc in moves:

        if b == 1:

            new = (m-dm, c-dc, 0)

        else:

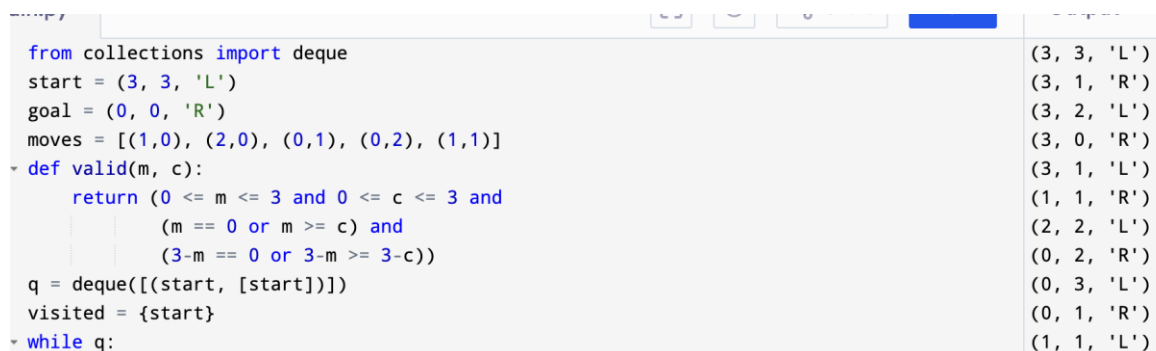
            new = (m+dm, c+dc, 1)

        if valid(new):

            queue.append((new, path+[new]))

```

Output:



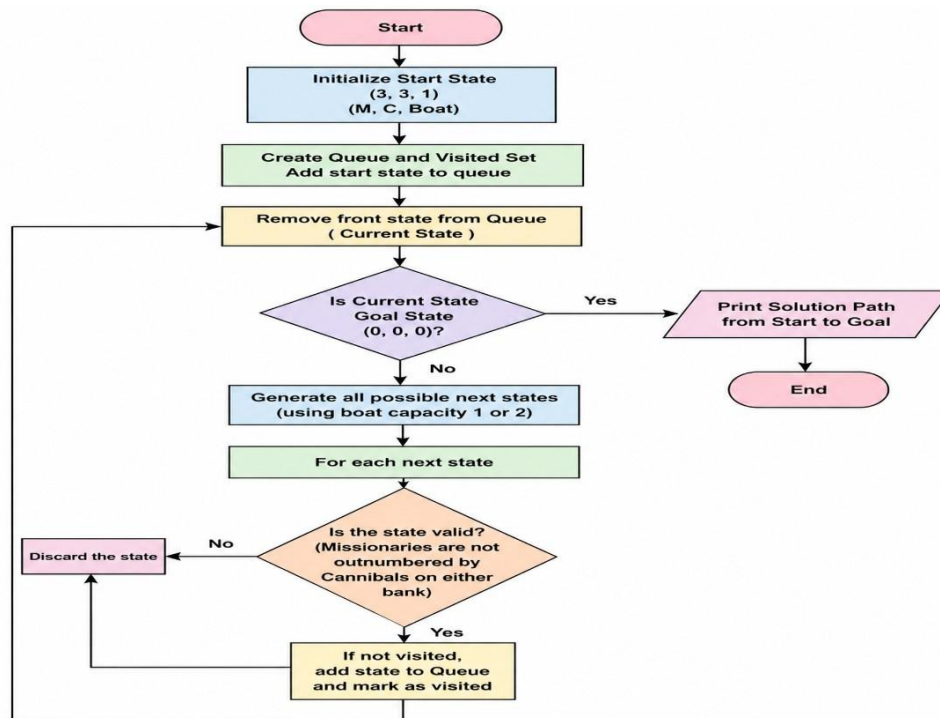
```

from collections import deque
start = (3, 3, 'L')
goal = (0, 0, 'R')
moves = [(1,0), (2,0), (0,1), (0,2), (1,1)]
def valid(m, c):
    return (0 <= m <= 3 and 0 <= c <= 3 and
            (m == 0 or m >= c) and
            (3-m == 0 or 3-m >= 3-c))
q = deque([(start, [start])])
visited = {start}
while q:

```

(3, 3, 'L')
 (3, 1, 'R')
 (3, 2, 'L')
 (3, 0, 'R')
 (3, 1, 'L')
 (1, 1, 'R')
 (2, 2, 'L')
 (0, 2, 'R')
 (0, 3, 'L')
 (0, 1, 'R')
 (1, 1, 'L')

Flowchart:



Result: Hence, Missionaries and Cannibals problem has solved